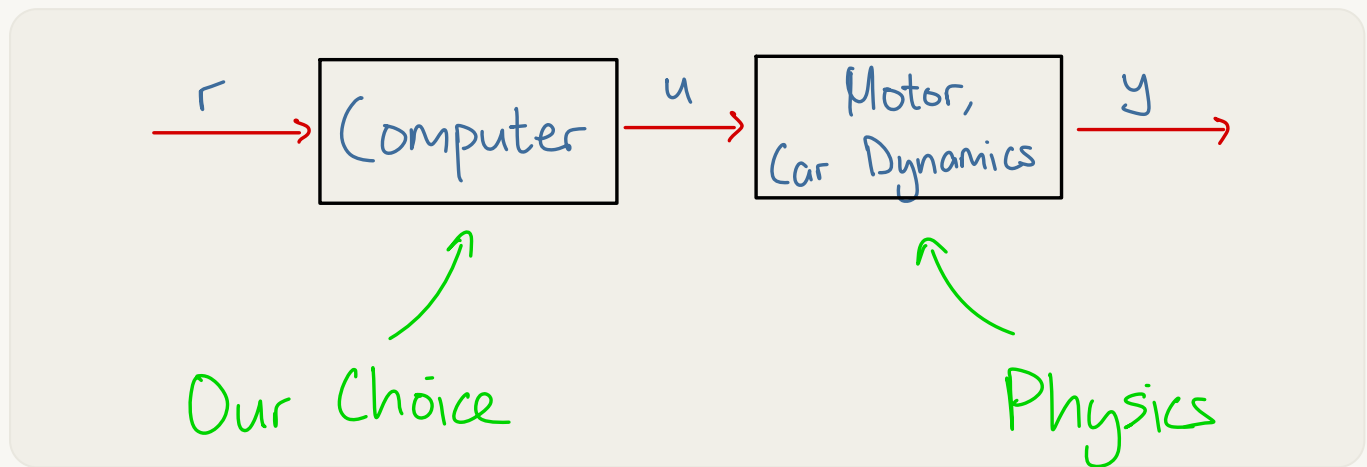




Where did we leave off?



r : Reference · u : Input · y : Output

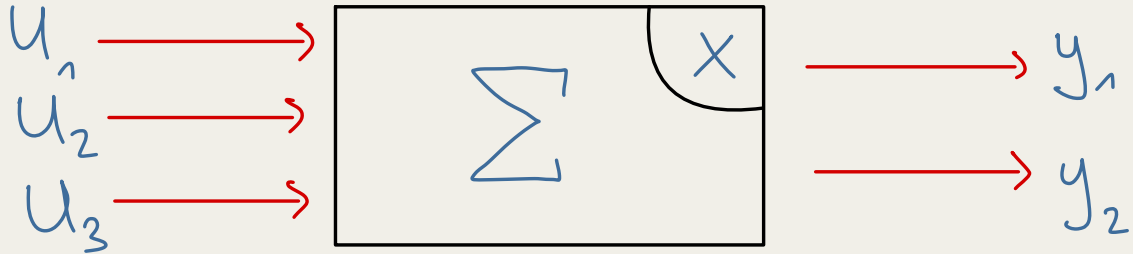
⇒ To find a controller, we must first find out how the plant works! ⇒ Make a model of it!

So how do we find this model?

⇒ Use equations to describe the dynamical behaviour of a system.

- system **evolves over time**
- system **has inputs and outputs**

The general system



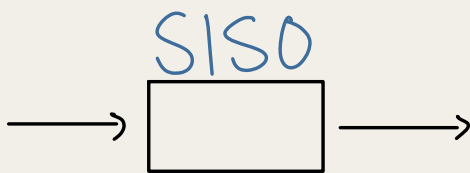
u_i — inputs

y_i — outputs

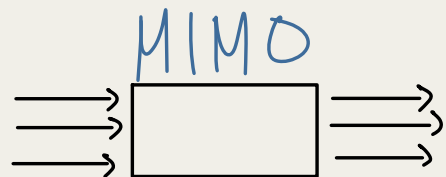
Σ — diff' eqns of the system

x — **state variables**

Note: the general system can have multiple in-/outputs, and the #inputs can differ from the #outputs.



Single input, single output
focus of this course

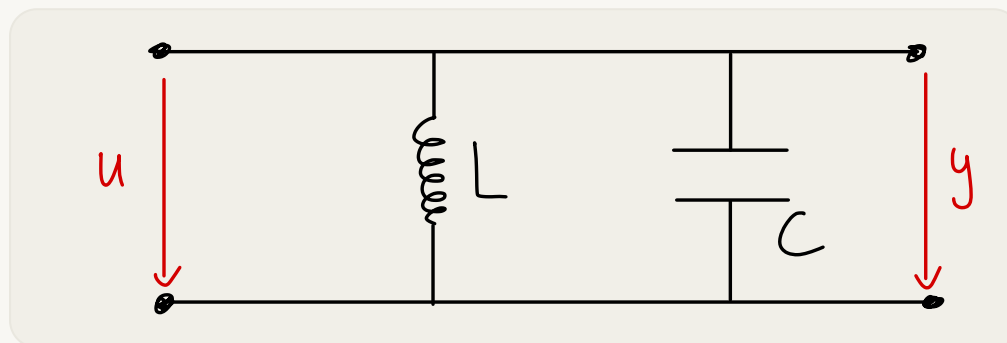


Multiple inputs, multiple outputs
focus of CS2

But what are state variables x ?

Def — internal variable quantity. The output is **affected** by the states. They are not always directly **visible** if you only look at the input and output.

Example: RC-circuit



u : input voltage [V]
 y : output voltage [V]
 L : inductor [H]
 C : capacitor [F]

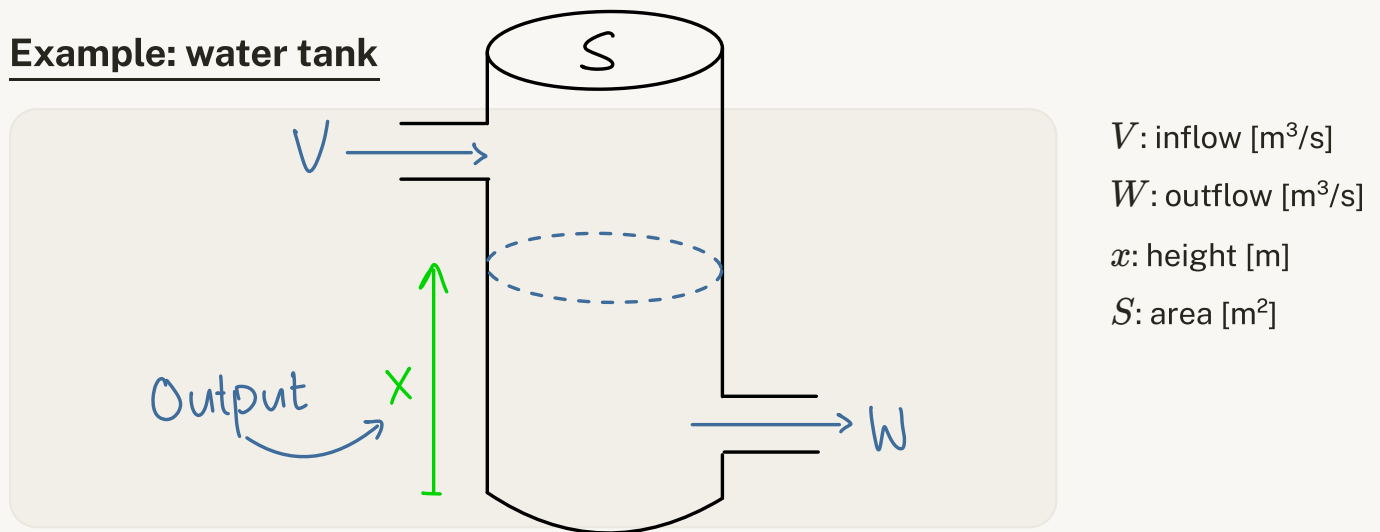
⇒ Output y is a function of the **E-field (state)** at the capacitor and the **B-field (state)** at the inductor (and the input u).

$$y = f(\underbrace{E, B}_{\text{states}}, u)$$

We do not see the states in the output, but the output is influenced by them!

In many cases, the output is just one of the states, which simplifies the calculation.

Let's look at an example from the lecture.

Example: water tank

Now we can use a common modeling technique called the **reservoir** based approach:

$$\frac{d}{dt} \text{storage} = \sum \text{inflows} - \sum \text{outflows}$$

Think of the derivative as a “change”.

Here: storage $\triangleq$ water volume $\triangleq S \cdot x$

$$\begin{aligned} \frac{d}{dt}(S \cdot x) &= V - W \\ y &= x \end{aligned}$$

Simplifying:

$$\begin{aligned} \dot{x} &= \frac{1}{S}(V - W) \\ y &= x \end{aligned}$$

$\Rightarrow$ Any model will take this form!

General form of diff' eqns:

$$\dot{x}(t) = f(x(t), u(t)) \quad (\text{I})$$

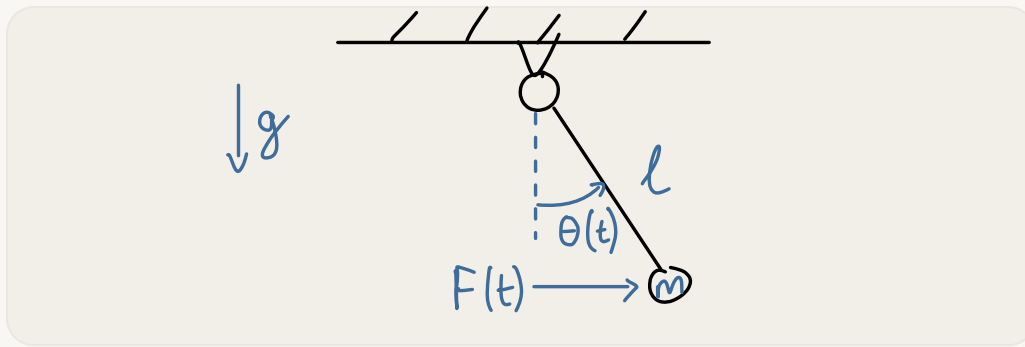
$$y(t) = h(x(t), u(t)) \quad (\text{II})$$

(I): internal system dynamics · (II): system observation

Note: for a given system the eqns are not unique! We can measure multiple things on one system! Ex: car output $\triangleq v, \dot{v}, \dots$

But the notation above only allows for first-order derivatives... this can't describe any system, right?

⇒ True! But there's a trick.

Example: pendulum

$l, m, g \triangleq$ constants

$F_p \triangleq$ input

$\theta \triangleq$ output

1) Describe dynamics

$\Rightarrow$ Conservation of angular momentum

$$\text{Inertia} \rightarrow J\ddot{\theta} = \sum M \leftarrow \text{Torques}$$

$$ml^2\ddot{\theta}(t) = -lmg \sin(\theta(t)) + l \cos(\theta(t))F_p(t)$$

$$y(t) = \theta(t)$$

How do we achieve this form?

$$\dot{x}(t) = f(x(t), u(t))$$

$$y(t) = h(x(t), u(t))$$

2) Introduce new state

Rename: $\theta(t) = \theta_1(t)$

Define: $\dot{\theta}(t) = \theta_2(t)$

$\Rightarrow$ We deduce: $\dot{\theta}_1(t) = \theta_2(t)$

3) Write total system

$$\begin{aligned}\dot{\theta}_1(t) &= \theta_2(t) \\ m l \ddot{\theta}_2(t) &= -l m g \sin(\theta_1(t)) + l \cos(\theta_1(t)) F_p(t) \\ y(t) &= \theta_1(t)\end{aligned}$$

$\theta_1(t), \theta_2(t)$ are the **states** $x_i(t)$

$F_p(t)$ is the **input** $u(t)$

$y(t)$ is the **output** $y(t)$

4) Write in general form

$$\begin{aligned}\dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -\frac{g}{l} \sin(x_1(t)) + \frac{1}{ml} \cos(x_1(t)) u(t) \\ y(t) &= x_1(t)\end{aligned}$$

We have achieved this form!

$$\dot{x}(t) = f(x(t), u(t))$$

$$y(t) = h(x(t), u(t))$$

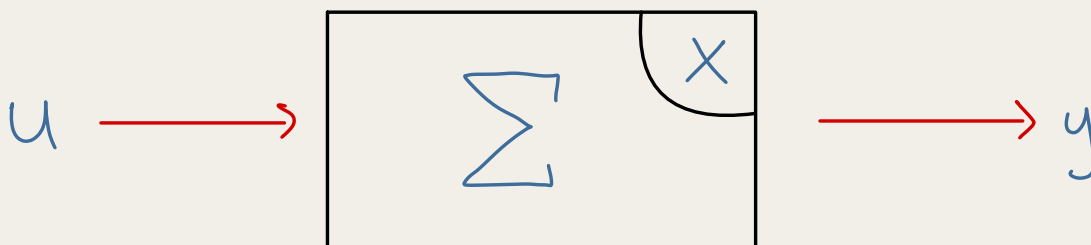
Intermediate quiz questions

Q1) Conservation of angular momentum is a reservoir-based modeling approach. **True** / False

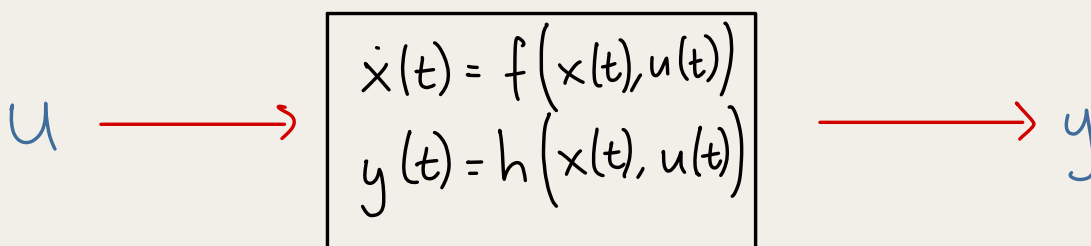
Q2) Inputs and outputs are time-dependent, but a general system is not. True / **False**

Q3) The output of a system is independent of the chosen sensor. True / **False**

How does this fit into the bigger picture?



We now know what this truly is:



Next week we will see what it means to **linearize** a system. Once we do that, we can write the signal–system relationship as such:

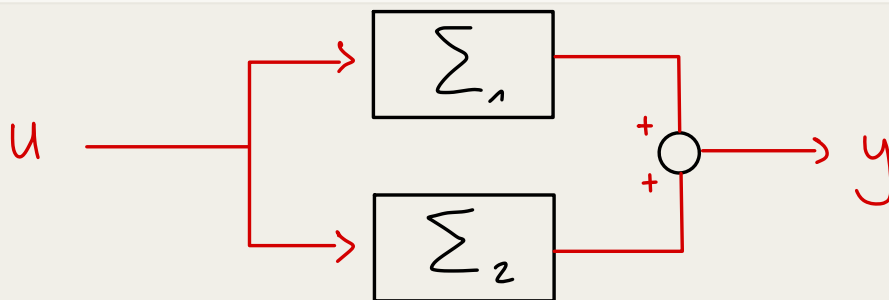
$$y = \Sigma u$$

This will allow us to analyze more complex interconnections of systems.

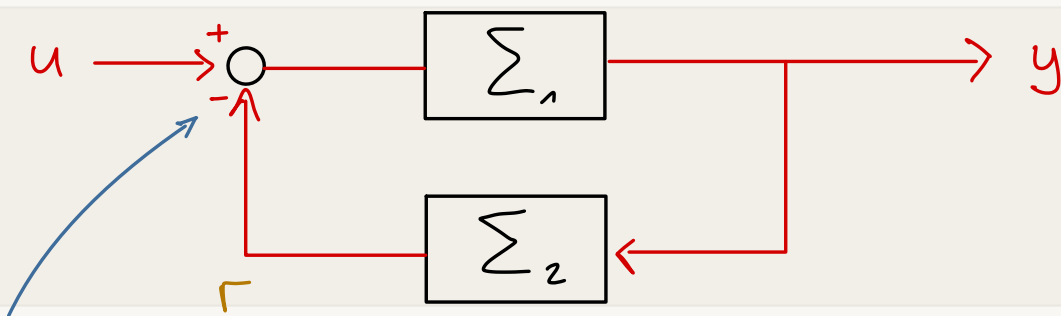
In essence, any interconnection can be broken down into three fundamental connection types.

1) Serial interconnection

$$\Rightarrow y = \Sigma_2 \Sigma_1 u$$

2) Parallel interconnection

$$\Rightarrow y = (\Sigma_1 + \Sigma_2)u$$

3) Feedback interconnection

Note: this defines if we have negative or positive feedback.

$$\Rightarrow y = \Sigma_1(u - r) \quad (I)$$

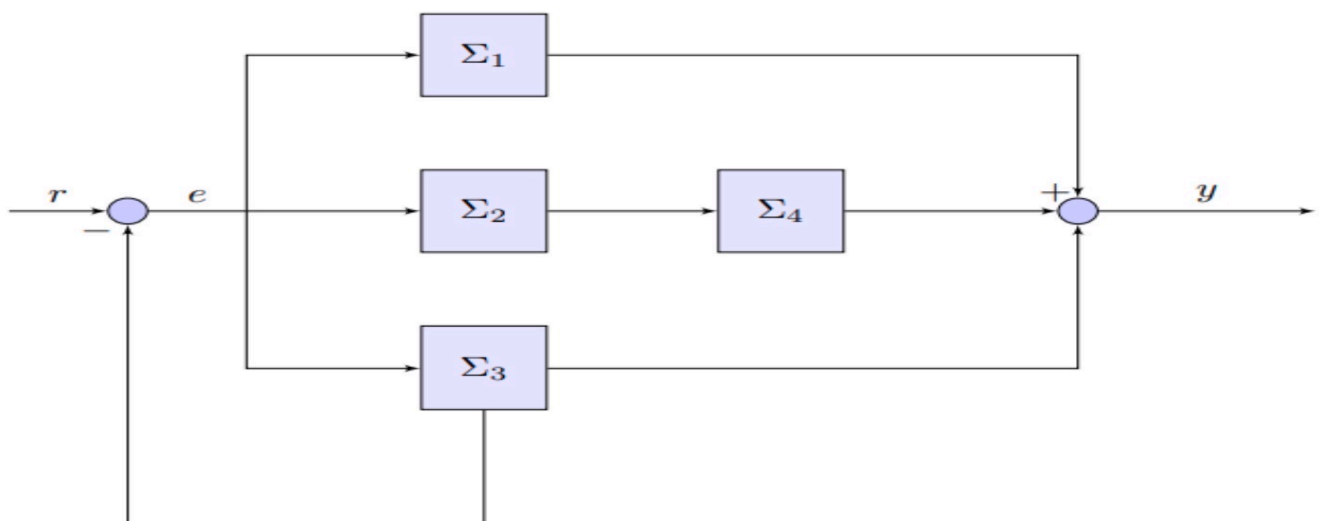
$$r = \Sigma_2 y \quad (II)$$

Combining (I) and (II):

$$\begin{aligned}
 y &= \Sigma_1(u - \Sigma_2 y) \\
 &= \Sigma_1 u - \Sigma_1 \Sigma_2 y
 \end{aligned}$$

Solve for y :

$$\begin{aligned}
 y(1 + \Sigma_1 \Sigma_2) &= \Sigma_1 u \\
 y &= (1 + \Sigma_1 \Sigma_2)^{-1} \Sigma_1 u
 \end{aligned}$$

Exam 2018*Question 5 — choose the correct answer. (1 point)*

You are given the above system diagram. What is the associated transfer function from $r \rightarrow y$?

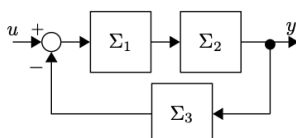
A $\Sigma_{r \rightarrow y} = (\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3)$

C $\Sigma_{r \rightarrow y} = \frac{\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3}{1 + \Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3}$

B $\Sigma_{r \rightarrow y} = \frac{\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3}{1 + \Sigma_2 \Sigma_4}$

D $\Sigma_{r \rightarrow y} = \frac{\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3}{1 + \Sigma_3}$

From Summary Sheet:

Modeling

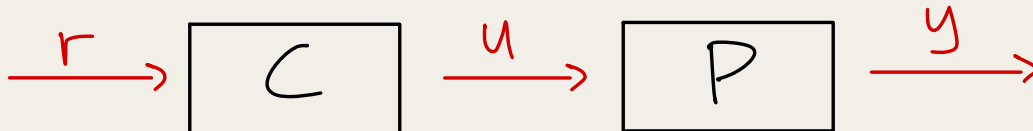
$$y = \frac{\Sigma_2 \Sigma_1}{1 + \Sigma_2 \Sigma_1 \Sigma_3} u$$

Figure 1: Negative feedback loop.

How do we now apply this to the structure we initially wanted to control?

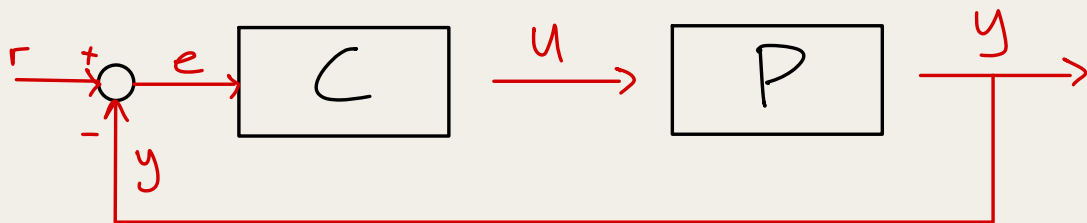
We had the following signals/systems:

r : reference · u : input · y : output C : controller · P : plant



$$y = Pu, \quad u = Cr$$

$$\Rightarrow y = PCr \quad (\text{feed-forward})$$



$$y = Pu, \quad u = Ce, \quad e = r - y$$

$$\Rightarrow u = C(r - y)$$

$$\Rightarrow y = PC(r - y) \quad (\text{feedback})$$

$$y(1 + PC) = PCr$$

$$y = (1 + PC)^{-1} PCr$$

Why is feedback so crucial?

⇒ Feedforward only works if we have perfect knowledge of the plant in a perfect world.

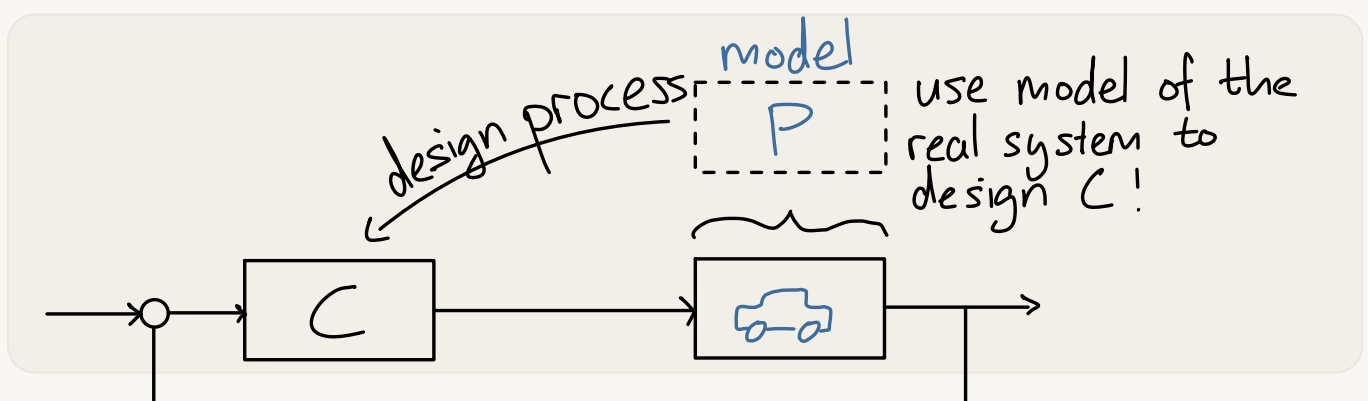
With feedback, we can:

- + stabilize an unstable system
- + compensate modeling uncertainties
- + reject external disturbances

But feedback can also:

- introduce instability even in an otherwise stable system
- feed noise into the system

Before we get to the quiz questions, one final sidenote – what is the relation between plant and model?



Use model P of the real system to design C .

Q4) It is possible and a good idea to combine feedback and feedforward.

True / False

Q5) We treat system blockcharts just like resistances in a circuit. True

/ **False**

Q6) When transforming a general system to a first-order representation, we inevitably lose information. True / **False**

Thanks for coming, have a wonderful weekend!

- ▷ inputs for summary sheet on moodle
- ▷ Problem Set 01 : Normal Tasks or Robot Task
- ▷ sections 1,1 / 1,2 / 1,3 / 1,7 from lecture notes